AARON KYLE VILLEGAS

Actuarial Analyst | Passed SOA Exam P | Python, SQL

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PROFESSIONAL SUMMARY

Applied and Computational Mathematics graduate (Data Science specialization) who applies probability and statistical modeling to large datasets and builds time-series forecasts. Passed SOA Exam P (September 2026) and is preparing for Exam FM, backed by coursework in probability, fixed income and statistical methods. Uses Excel and Python to build financial and statistical models and explain findings clearly to non-technical audiences.

EDUCATION

University of California, Irvine June 2026

Bachelor of Science, Applied and Computational Mathematics (Data Science)

GPA: 3.911 | Completed B.S. in 3 Years (Graduated Early) | Dean's Honor List | Howard Tucker Award in Mathematics | Cum Laude

Actuarial Exams (SOA): Passed Exam P (September 2026) | Planning for Exam FM (Financial Mathematics) in December 2026

Relevant Coursework: Probability I & II (MATH 130A: A, MATH 130B: B+), Statistical Methods for Data I & II (STATS 110: A, STATS 111: A+), Financial Derivatives & Fixed Income (MATH 134B: A+, MATH 134A: A+), Optimization I & II (MATH 110A: A, MATH 110B: A+)

TECHNICAL SKILLS

Core Analytics Tools: Excel, SQL (PostgreSQL, BigQuery), Python, R (coursework), Tableau

Statistical Libraries: Pandas, statsmodels

Domain & Operational Methods: Generalized Linear Models (coursework), Time-Series Forecasting, Monte Carlo Simulation, Statistical Analysis, Predictive Modeling

RELEVANT EXPERIENCE

Humn Solutions | *Software Developer Intern* September 2026 - Present | Remote

- Built an automated data pipeline using Python and Supabase to collect, rank, and summarize daily data from 13+ public sources.

Handshake AI | *AI Benchmark Engineering Fellow* July 2026 - August 2026 | Remote

- Developed automated pytest verification suites applying baseline trend modeling and median aggregation to evaluate model outputs against strict requirements, saving 3 to 4 engineering hours per incident by replacing manual debugging.

Jane Street | *Academy of Math and Programming (AMP) Scholar* June 2023 - July 2023 | NYC

- Selected for Jane Street's competitive Academy of Math and Programming after rigorous quantitative screening; awarded $5,000 scholarship.
- Developed and optimized a Python-based trading strategy with a team in a simulated environment, finishing in the top 25%.

QUANTITATIVE & ANALYTICS PROJECTS

Macroeconomic & Climate Analytics Platform (Macro Data Pipeline) April 2026 - Present

Python, SQL, PostgreSQL, BigQuery, statsmodels, SARIMAX

- Built an ETL/ELT pipeline in Python that independently sourced and structured 9M+ time-series records from multiple APIs (FRED, Open-Meteo) into PostgreSQL and BigQuery for multi-variable analysis.
- Engineered a SARIMAX time-series forecasting engine (statsmodels) across economic indicator and retail sales domains, with RMSE, MAE, and MAPE forecast accuracy metrics.

Multi-Echelon Replenishment & Stockout Risk Simulator August 2026

Python, SciPy, DuckDB, Streamlit, Plotly

- Modeled multi-echelon inventory replenishment logic by calculating optimal Safety Stock and Reorder Points under demand and lead-time variability using SciPy Monte Carlo simulations.

Fixed-Income & Portfolio Modeling May 2026

Microsoft Excel

- Constructed dynamic Excel models applying Markowitz portfolio theory to optimize asset allocation, applying the one-fund theorem to evaluate expected returns.
- Evaluated fixed-income securities by modeling bond prices, cash flow streams, and yield to maturity, executing portfolio immunization and risk management strategies.

Deriving & Understanding the Black-Scholes Option Pricing Model December 2025

Python, LaTeX, Stochastic Calculus, Quantitative Finance

- Authored an in-depth mathematical paper deriving the Black-Scholes PDE using Itô's Lemma, Geometric Brownian Motion (GBM), and dynamic delta hedging to eliminate portfolio variance under no-arbitrage principles.
- Implemented a Monte Carlo simulation in Python to test and benchmark the Black-Scholes pricing model against simulated asset price paths.